

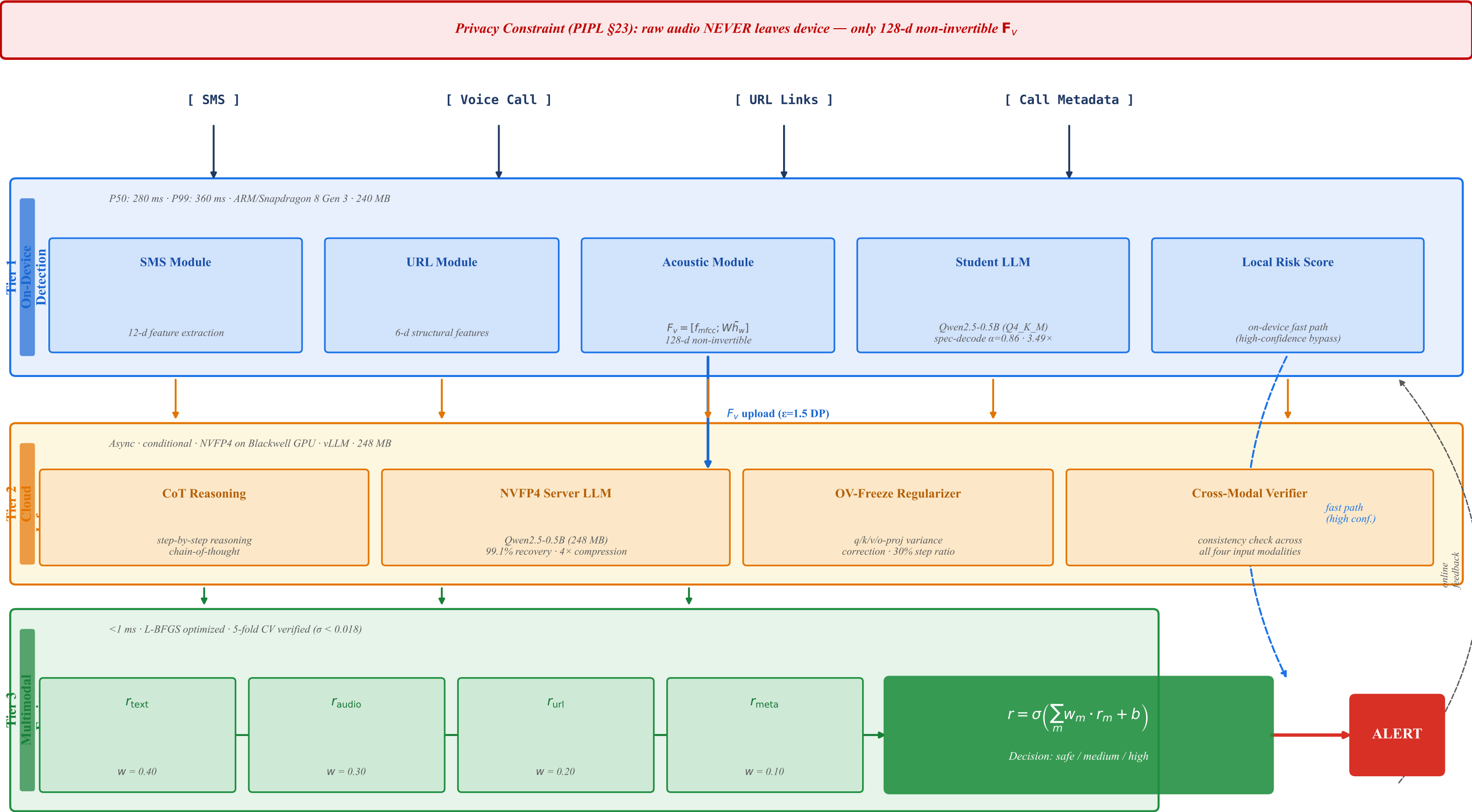


Fig. 1. QAD-MultiGuard three-tier edge-cloud collaborative architecture for telecom fraud detection. Tier 1 performs on-device detection with speculative decoding ($\alpha=0.86, \gamma=5, 3.49\times$ measured on Snapdragon 8 Gen 3). Tier 2 provides conditional cloud inference with NVFP4 QAD + OV-Freeze (99.1% F1 recovery). Tier 3 fuses all modalities via L-BFGS optimized σ -fusion ($\sigma < 0.018$ across 5-fold CV). Federated learning is future work (§5.2) and not part of the current Tier 2.